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Predicting Cowpea Yield with Machine Learning

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Introduction

- **Brief Background / Context:**

Cowpea is an important legume crop widely grown for its nutritional value and ability to fix nitrogen, but its yield depends on proper nutrient management.

- **Agriculture Problem Area:**

Farmers often face difficulty in determining the right amount of nitrogen fertilizer, leading to low yield or excessive input use.

- **Importance of Data-Driven Approaches:**

Machine learning helps analyze complex relationships between inputs and yield, enabling accurate predictions and better decision-making in

Problem Statement

- Predict cowpea crop yield using machine learning techniques
- • Analyze the effect of nitrogen application and plant growth parameters on yield
- • Develop accurate and reliable prediction models
- • Compare different algorithms to identify the best-performing model
- • Scope limited to a small experimental dataset with selected features

Motivation

- Crop yield prediction is important for improving food production and farm efficiency
- • Helps farmers determine optimal nitrogen usage and reduce input costs
- • Supports better decision-making in crop management and resource allocation
- • Useful in precision agriculture and smart farming practices
- • Contributes to research and development in agricultural machine learning
- • Promotes sustainable farming by reducing fertilizer overuse and environmental impact

Objectives of the Project

- Gain knowledge in machine learning, data analysis, and data visualization
- • Understand model training, evaluation, and performance metrics
- • Identify patterns and relationships between input features and yield
- • Generate accurate predictions and meaningful insights for farming decisions
- Compare different algorithms to identify the best-performing model
- Develop machine learning models to predict cowpea crop yield

Dataset Overview

Dataset Source: Experimental agronomic dataset (field experiment on cowpea)

- **Number of Records and Features:**

20 samples

3 input features + 1 target variable

- **Types of Features:**

Numerical features (nitrogen level, plant height, leaf area, yield)

Crop and growth-related parameters

- **Preprocessing Steps:**

Checked for missing values (none present)

Selected relevant input features

Removed categorical variables (treatment, replication)

Train-test split (80% training, 20% testing)

Methodology / Approach

Data Cleaning & Preprocessing

- Exploratory Data Analysis (EDA)
- Model / Algorithm Selection (Linear Regression, Decision Tree, Random Forest, ANN)
- Model Training and Validation (80% training, 20% testing)
- Evaluation Metrics: R^2 Score, MAE, RMSE

Workflow diagram illustrating the pipeline

- Data loading → Feature selection → 80/20 train-test split
- (random_state=42) → Model training → Evaluation with R^2, MAE, RMSE.

Fig. 1: Methodology Flowchart for Cowpea Yield Prediction

Tools and Technologies Used

- **Libraries Used:**
- scikit-learn
- pandas
- numpy
- matplotlib
- seaborn
- **• Development Environment**:
- Jupyter Notebook
- Google Colab
- **Programming Language**:
- Python

Results

- Model performance was evaluated using R^2 Score, MAE, and RMSE
- **Random forest** achieved the best performance
- R^2 Score = 0.9931
- MAE = 18.54 kg/ha
- RMSE = 19.48 kg/ha

Decision tree also performed well with high accuracy and better generalization

R^2 Score = 0.9898

- Linear Regression showed good prediction accuracy but could not capture non-linear trends
- ANN gave poor performance due to small dataset size and lack of normalization

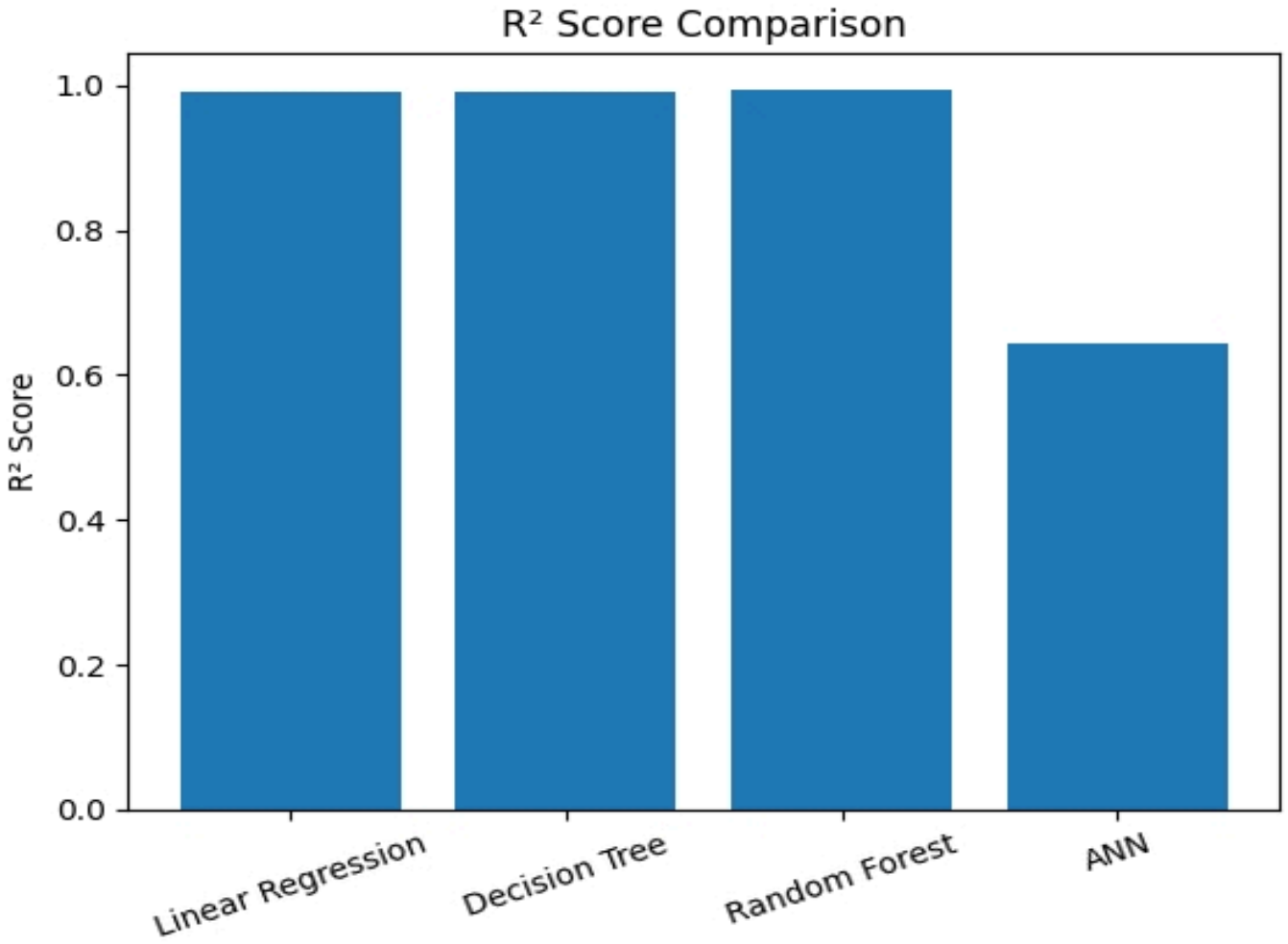

Fig.2 MAE Comparison Bar Chart — Compares Mean Absolute Error across all four models.
Random Forest and Decision Tree bars are shortest (MAE=9.71), while ANN bar is tallest

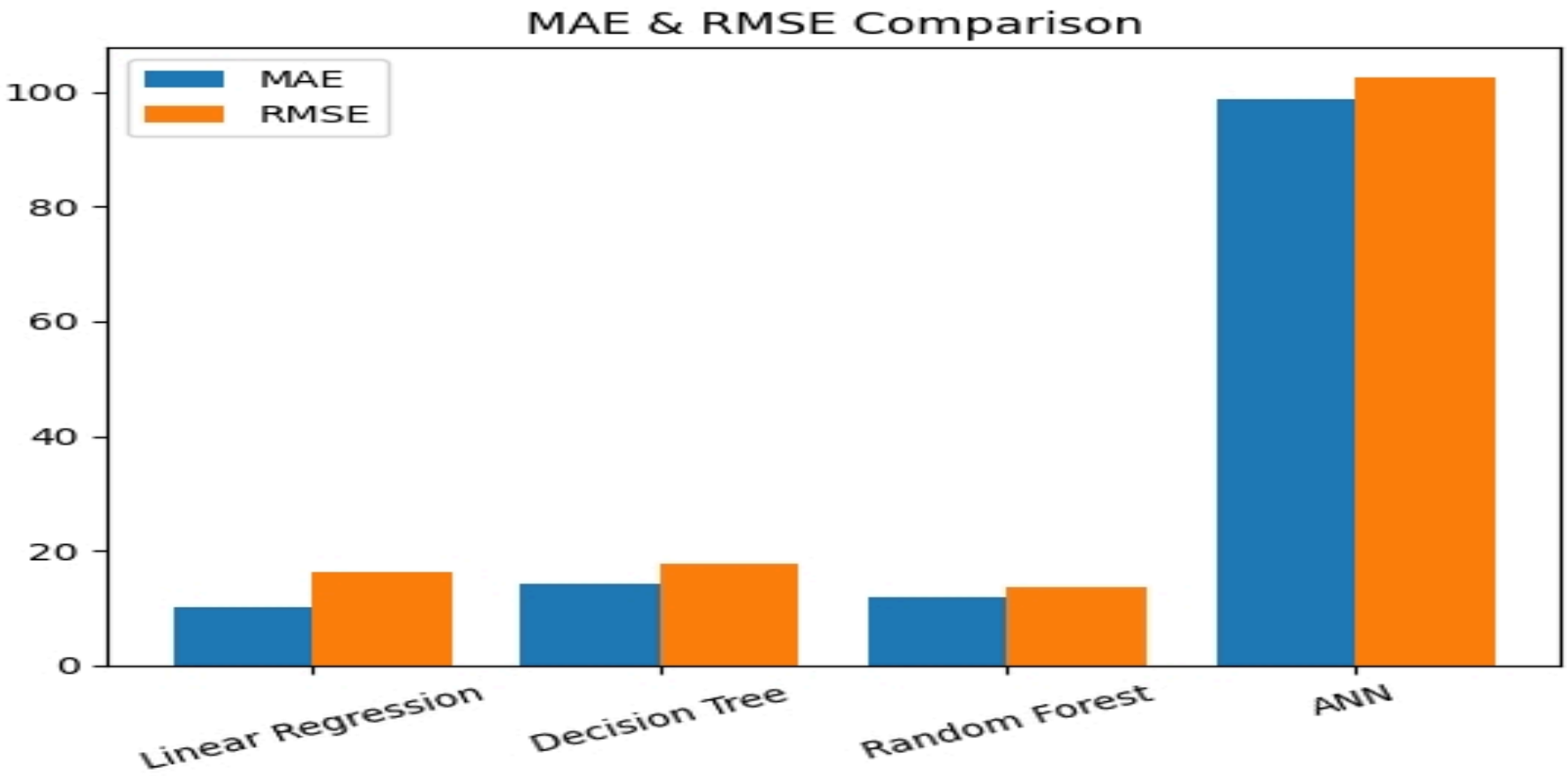

Fig..3 R^2 Score Comparison Bar Chart — All models except ANN exceed R^2=0.99, with RF scoring highest at 0.9953. ANN's negative R^2 confirms it performs worse than a simple mean predictor on this dataset.

7.4 Random Forest- Hyperparameter Tuning

Table 8: Different n_estimators Comparison (MAE, MSE, RMSE, R^2 Score)

Sample No.	n_estimators	MAE	MSE	RMSE	R2 score
1	10	14.0	229.375	15.145	0.9922
2	50	11.15	180.375	13.430	0.9938
3	100	9.712	139.57	11.814	0.995
4	200	10.899	170.438	13.055	0.9942
5	300	12.075	197.372	14.048	0.9933

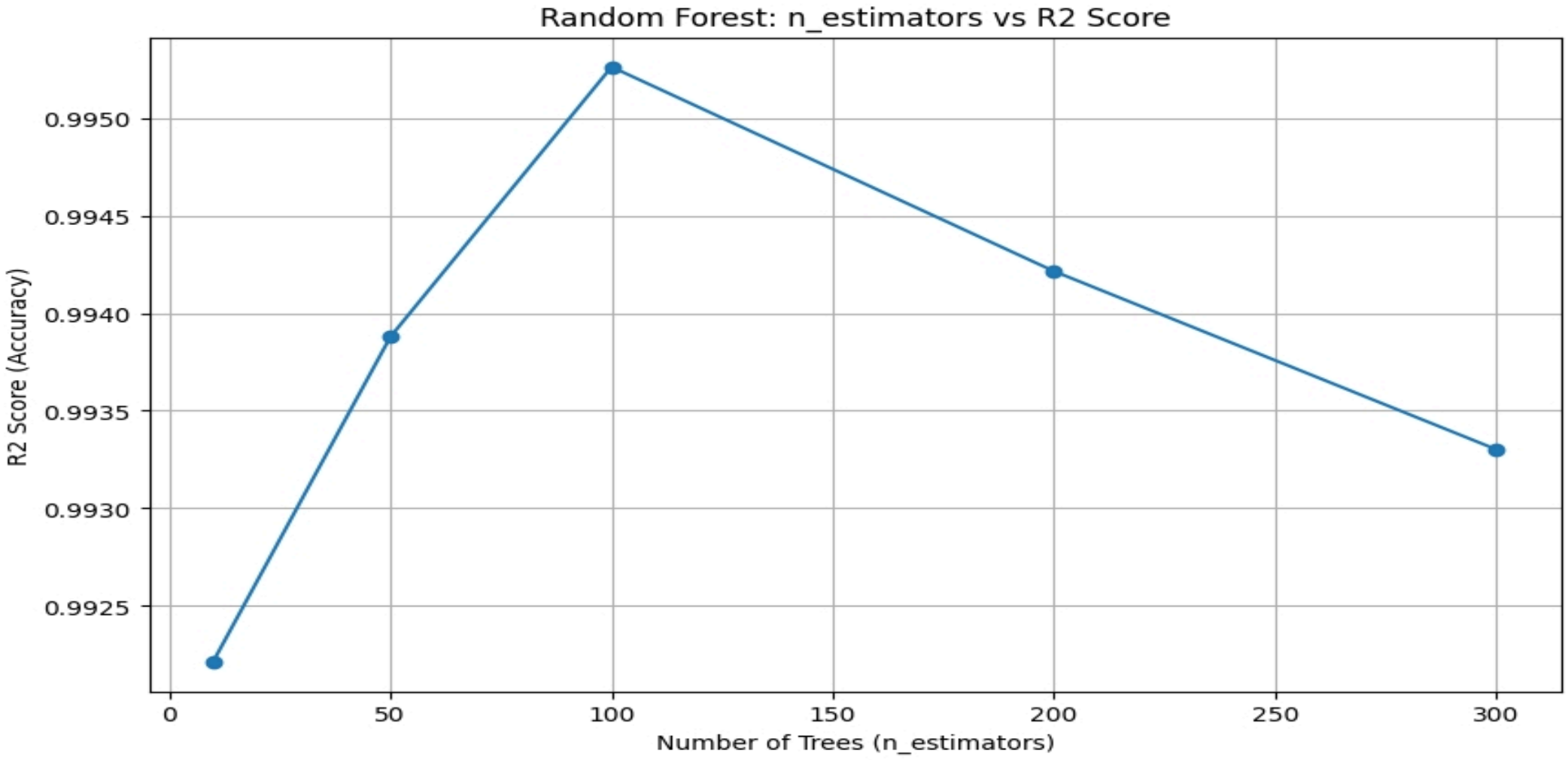

Fig 4:Random Forest:n_estimatores vs R2 Score

Visualizations Used:

R^2 comparison bar graph

MAE and RMSE comparison graph

Actual vs Predicted Yield plots

Nitrogen vs Yield curve

Decision Tree structure and feature importance

Observations:

Nitrogen level and leaf area are the most important factors affecting yield

Yield increased with nitrogen up to 60 kg/ha and decreased at 80 kg/ha

Tree-based models captured the non-linear relationship more effectively than ANN and Linear Regression

Discussion / Analysis

- Nitrogen application and leaf area are the most influential factors affecting cowpea yield
- Yield increases up to an optimal nitrogen level (60 kg/ha) and then declines
- Tree-based models (Decision Tree, Random Forest) perform best for this dataset

Challenges Faced:

Small dataset size (only 20 samples)

Limited features considered for prediction

ANN model did not converge properly

Lack of feature scaling and hyperparameter tuning

- **Limitations**:

Dataset does not include factors like soil type, climate, irrigation, pests

No cross-validation due to limited data

Results may not generalize to larger or real-world datasets

Impact on Agriculture:

Helps farmers optimize nitrogen usage for better yield

Supports data-driven decision-making in farming

Useful for precision agriculture and sustainable farming practices

Conclusion and Future Work

- Machine learning models were successfully used to predict cowpea yield
- Decision Tree and Random Forest showed highest accuracy and reliability
- Optimal nitrogen level identified as 60 kg/ha for maximum yield
- Leaf area and nitrogen are key factors influencing yield

Future Work:
Use larger and more diverse datasets **(multi-location, multi-season)**
Include additional features like **soil type, weather, and irrigation**
Apply advanced models (Deep Learning, ensemble techniques)
Develop **real-time prediction** systems for farmers

Practical Applications:
Helps farmers **optimize fertilizer usage**
Supports **precision agriculture and smart farming**
Improves **yield prediction and decision-making**

References / Acknowledgements

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